II Math Analysis

1. Real Number and Complex Numbers

1.1 Completeness of $\mathbb{R}$: Def. Least upper bound property of $\mathbb{R}$

for any $E \subset \mathbb{R}$, if $\exists m$ s.t. $\forall x \in E$, $x \le m$, then $\exists \sup E \in \mathbb{R}$ / Cauchy Convergence

1.2 $\mathbb{R}, \mathbb{C}$ are metric spaces, complete, and $\mathbb{R}$ is ordered

So there are some topological concepts: open, closed, open neighborhood, convergence, etc.

2. Sequence and Series

2.1 $(x_n) \subset \mathbb{R}/\mathbb{C}$ $x_n \to x_0$ if $\forall \varepsilon > 0$ $\exists N$, $\forall n \ge N$ $|x_n - x_0| < \varepsilon$

2.2 If $x_n \uparrow$, bounded, then $\exists \lim x_n = \sup\{x_1, x_2 \ldots\}$

2.3 Criterion for convergence: monotone + bounded
sandwich

2.4 Convergence is preserved for $+, -, \times, \div$ (/ eliminated)

~~$\lim_{n\to\infty}$ an a function~~ ~~$\lim_{n\to\infty}$ $\mathbb{R}^\infty \to \mathbb{R}$ not all~~

$\lim_{n\to\infty}$: Cauchy Sequences $\to \mathbb{R}$ that is closed under $+ - \times \div$

2.5 $\lim_{n\to\infty}$ is monotone, where $(x_n) \le (y_n)$ means that for large n

$x_n \le y_n$ $\forall n$ sufficiently large

3 Series $\sum_{n=1}^{\infty} x_n$

3.1 Def. $S_N = \sum_{n=1}^{N} x_n$, If S_N is Cauchy then $\lim_N S_N$ is $\sum_{n=1}^{\infty} x_n$

3.2 Nonnegative terms if $\forall n$, $x_n \ge 0$, then we have criterion for convergence

1. Ratio test
2. Root test
3. Comparison test

3.3 $\sum x_n + \sum y_n$, $\sum x_n y_{m-n}$ product of two series (convolution), double summation $\sum\sum x_{nm}$

3.4 Absolute and conditional convergence

<u>Thm.</u> $\sum |x_n| < \infty$, $\sum x_n$ converges $\left|\sum_N^M x_n\right| \le \sum_N^M |x_n| \to 0$

3.5 Riemann Rearrangement Theorem

~~4 Differentiation~~

More on Series, $\sum_{i=1}^{\infty} : E \subset \mathbb{R}^{\mathbb{N}} \to \mathbb{R} \cup \{+\infty\}$

E S_N is Cauchy, Σ is also monotone

$\mathbb{R}_+^N \subset E$ although it ~~can~~ can be

4 Differentiation / Continuity

4.1 Let $f: \mathbb{R} / \mathbb{C} \to \mathbb{C}$, then f is continuous if $\forall$ open $G \subset \mathbb{C}$ $f^{-1}(G)$ is open

4.2 Continuity + compactness

+ connectedness

sequential continuity

4.3 Differentiation if $f(x) = f(x_0) + a(x - x_0) + o(|x - x_0|)$ as $x \to x_0$

we say a is the derivative of f at x_0, $f'(x_0)$

4.4. Meanvalue theorem

1.2. Leb measurability

RMK. Let's think about the measures we've encountered

(1) Elementary set with elementary measure (seems natural)

(2) Jordan measure: approximated by elementary sets

$$J^*(E) = \inf_{\substack{B:\text{elem} \\ E\subset B}} m(B) \qquad J_*(E) = \sup_{B\subset E} m(B)$$

Q
- (2.1) Definition
- (2.2) What sets are measurable
  - criteria for being measurable
  - what set operations preserve meas.
- (2.3) What's the measure of meas. sets
  - monotone?
  - additivity?
  - subadditivity?
  - translation invariance?
  - under linear transformation?
- (2.4) Uniqueness of measure
- (2.5.) how to calculate the measure

(3) Riemann Integral, (Darboux integral)

- (3.1) Definition, use what func. to approximate?
- (3.2) Integrability of functions
- (3.3) Properties of Integral
  - Linearity
  - monotonicity
  - Indicator (relation to the measure)

□

1.2.1 Leb outer measure

Def. $m^*(E)$, for any $E\subset\mathbb{R}^d$, $m^*(E)\in[0,\infty]$

$$m^*(E) = \inf_{\substack{(B_j)\text{ boxes} \\ E\subset\bigcup B_j}} \sum_{j=1}^{\infty} |B_j|$$

□

Q2.2 Now the domain of m^* is $2^{\mathbb{R}^d}$. That answers Q(2.2)

Q2.3 $m^*(E)$ is monotone, subadditive (countably)

$m^*(\phi)=0$

m^* is <u>not</u> additive, but additive for sets separated by dist >0

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On Measure theory of $\mathbb{R}^d$

Once we have some properties about the Lebesgue outer measure, we can continue to Leb measure

[1] Elementary Measure defined for elementary sets · finite union of boxes

$$m_E^*(E) = \sum_{i=1}^{n} |B|$$

[2] Jordan measure and Jordan measurable sets

[3] Jordan measure and Riemman Integral (Darboux integral)

[4] Lebesgue outer measure and then Leb measure on $\mathbb{R}^d$

<u>Def</u>. A box : $B = I_1 \times \cdots \times I_d$ $\quad I_1$ takes the form (a,b) $[a,b]$ $(a,b]$ or $[a,b)$

volume of a B is $|B| = \prod_{j=1}^{d} |I_j|$

<u>Def</u>. Jordan measure, a set $E \subset \mathbb{R}^d$ has Jordan outer measure

$$JM^*(E) := \inf_{\substack{B_1, \cdots B_n \\ E \subset \cup B_j}} \sum_{j=1}^{n} |B_j| \quad \text{and inner measure} \quad JM_*(E) = \sup_{\substack{B_1, \cdots, B_n \\ \cup B_j \subset E}} \sum |B_j|$$

If $JM^*(E) = J_*(E)$, then we say E is J-measurable

properties of Jordan measurable set

<u>prop1</u> If $E, F \in J\text{-meas}(\mathbb{R}^d)$, $E \cup F$ $\quad E \cap F$ $\quad E \setminus F$ are J-meas.

proof. [1] E has B

Lemma If $E \subset F$, then $JM^*(E) \leq JM^*(F)$

$JM^*(\phi) = 0$

$JM_* \leq JM^*$

Additivity

Related to integral under the graph

linear transformation

prop 1 $J(\phi)=0$ because ϕ has elementary measure 0

prop 2. If $E \subset F$ then $J^*(E) \leq J^*(F)$

proof If $J^*(F)=\infty$, obvious

If $J^*(F)<\infty$ then $\forall B_1, \cdots B_n$ covering F

$E \subset \bigcup^n B_j$ hence $J^*(E) \leq \sum |B_j|$ Hence $J^*(E) \leq \inf \sum^n |B_j|$

$\therefore J^*(E) \leq J^*(F)$ $\square$

prop 3 J-meas sets form a <u>Boolean-Algebra</u>

prop 4. $\forall E \subset \mathbb{R}^d$, $J_*(E) \leq J^*(E)$

Because $\forall (A_j)_{j=1}^n$ fix any $(B_k)_{k=1}^m$ s.t. $\bigcup A_j \subset E \subset \bigcup B_k$

$\sum |B_k| \geq \sum A_j$ so $\sum |B_k| \geq J_*(E)$ which is true $\forall (B_k)_{k=1}^m$ covering E

Hence $\inf \sum |B_k| \geq J_*(E)$

$\square$

~~proof J-meas($\mathbb{R}^d$) is Boolean algebra~~

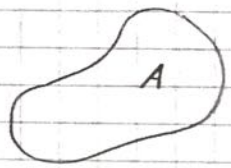
~~If $A_1, A_2 \subset \mathbb{R}^d$ $J^*(A_1 \cup A_2) \leq J^*(A_1) + J^*(A_2)$~~

$\boxed{\text{Approximation By elementary set}}$ $\forall A \in$ J-meas($\mathbb{R}^d$) $\Leftrightarrow \forall \varepsilon>0\ \exists B$ elementary,

$J^*(B \Delta A) < \varepsilon$

proof $\Rightarrow$ $m^*(A) = m_*(A)$ $\forall \varepsilon>0$ $\exists B_1, \cdots B_n$ s.t

$m^*(A) \geq \sum |B_j| - \frac{\varepsilon}{2}$



Then it holds

$\Leftarrow$ $m^*(A) \leq m^*(B) + \varepsilon$

$m_*(A) \geq m^*(B) - \varepsilon$

Let B be elementary, $B \Delta A = (A \setminus B) \cup (B \setminus A)$ which are disjoint

$J^*(B \Delta A) \leq J^*(A \setminus B) + J^*(B \setminus A)$

Jordan Measure: Approximation by Elementary

TFAE (1) $E \subset \mathbb{R}^d$ is J-meas

(2) $\exists A, B$ elementary, $\forall \varepsilon > 0$ $A \subset E \subset B$ $Jm(B \setminus A) < \varepsilon$

(3) $\forall \varepsilon > 0$ $\exists B$ elementary $Jm^*(E \Delta B) < \varepsilon$

proof (1) $\Rightarrow$ (2) If $E \in J(\mathbb{R}^d)$, then $m_*(E) = m^*(E)$

$\exists A, B$ s.t. $Jm_*(E) < Jm(A) + \frac{\varepsilon}{2}$

$Jm^*(E) > Jm^*(B) - \frac{\varepsilon}{2}$

$Jm(B) = Jm(B \setminus A) + Jm(A) < Jm(A) + \varepsilon$

$\Rightarrow$ $Jm(B \setminus A) < \varepsilon$

(2) $\Rightarrow$ (3) If $\forall \varepsilon > 0$ $\exists A, B$ elementary $A \subset E \subset B$ $Jm(B \setminus A) < \varepsilon$

Then $B \setminus E = B \Delta E \subset B \setminus A$

$Jm^*(B \Delta E) \leq Jm(B \setminus A) < \varepsilon$

(3) $\Rightarrow$ (1) If E: $\forall \varepsilon > 0$ $\exists B$ elementary $Jm^*(E \Delta B) < \varepsilon$

then $\cancel{Jm^*(E) \leq Jm^*(B) + Jm^*(E \Delta B) < Jm(B) + \varepsilon}$

$\cancel{Jm_*(E) = \sup_{A \text{ elementary}} Jm(A)}$

$\cancel{E \supset A}$

we don't have subadditivity yet

$\forall \varepsilon > 0$ $\exists B$ elementary $Jm^*(E \Delta B) = \inf_{\substack{B' \\ E \Delta B \subset B'}} m(B') < \varepsilon$

then $\exists B'$ s.t. $E \Delta B \subset B'$ s.t. $m(B') < 2\varepsilon$

Then $E \subset B \cup B'$ and $B \setminus B' \subset E$

B, B' disjoint

Because $E \subset B \cup (E \Delta B)$ and $B \setminus (B \setminus E) \supset B \setminus E \Delta B \supset B \setminus B'$

$\subset B \setminus E$

$E \Delta B \subset E \setminus B$

and so $Jm^*(E) \leq Jm^*(B) + Jm^*(B') = Jm^*(B) + 2\varepsilon$

and $\cancel{m^*(E)}$ $Jm_*(E) \geq m^*(B) - m^*(B') = m^*(B) - 2\varepsilon$

Hence $m^*(E) - m_*(E) \leq 4\varepsilon$

By arbitrary ε, $m^*(E) = m_*(E)$

$\blacksquare$

1.2.1 Properties of Lebesgue Outer Measure m^*

* **Prop** (1) $m^*(\phi)=0$ (2) $E\subset F\subset \mathbb{R}^d$, $m^*(E)\le m^*(F)$ (3)

(3) If $E_1, E_2 \cdots \subset \mathbb{R}^d$ is a countable seq of sets then

$$m^*(\cup E_n) \le \sum m^*(E_n)$$

proof. Only (3) is nontrivial. Let E_n has outer measure $m^*(E_n)$

$$m^*(E_n) = \inf_{E_n\subset \cup B_{n,k}} \sum |B_{n,k}|$$ suppose $\sum m^*(E_n) < \infty$. then

$\forall \varepsilon>0$ $\exists (B_{n,k})_{k=1}^{\infty}$ $E_n \subset \cup B_{n,k}$ and $m^*(E_n) \ge \sum |B_{n,k}| - \frac{\varepsilon}{2^n}$

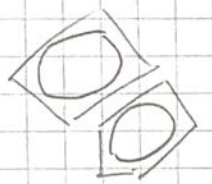
Then $\bigcup_{n=1}^{\infty}\bigcup_{k=1}^{\infty} B_{n,k} \supset \cup E_n$ and so

$$m^*(\cup E_n) \le \sum_{n,k} |B_{n,k}| = \sum_n \sum_k |B_{n,k}| \le \sum_n m^*(E_n) + \varepsilon$$

By ε is arbitrary, we have $m^*(\cup E_n) \le \sum m^*(E_n) + \varepsilon$ $\square$

Lemma 1.2.5 Let $E, F \subset \mathbb{R}^d$ be such that $\operatorname{dist}(E,F) = \inf\{|x-y| : x\in E, y\in F\} > 0$

Then $m^*(E\cup F) = m^*(E) + m^*(F)$



proof. From subadditivity $m^*(E\cup F) \le m^*(E) + m^*(F)$

To show $m^*(E) + m^*(F) \le m^*(E\cup F)$ if $E\cup F$ has ∞ outer measure then we are done.

Otherwise, let $\varepsilon>0$, $\exists (B_n)$ of boxes such that

$$\sum |B_n| \le m^*(E\cup F) + \varepsilon$$

$\forall r>0$, for any box B_n, B_n can be partitioned into a finite number of small boxes S_j each has $\sup\{|x-y| : x,y\in S_j\} < r$

And $|B_n| = \sum_{j=1}^{J} |S_j|$

Then Any box $B_n \cap E \ne \phi \Rightarrow B_n \cap F = \phi$

so $\cup B_n = \cup B_{1,j} \cup B_{2,j}$ $\quad E\subset \cup B_{1,j}$, $F \subset \cup B_{2j}$

$m^*(E) \le \sum_{j=1}^{\infty} |B_{1,j}|$ $\quad m^*(F) \le \sum_{j=1}^{\infty} |B_{2j}|$

By the finiteness of m^*

Exercise 1.2.4 $E, F \subset \mathbb{R}^d$, $E \cap F = \phi$. E, F closed, E is compact

show $\text{dist}(E, F) > 0$

proof. $\boxed{\begin{array}{l} \text{dist}(E,F) = \inf\{|x-y| : x \in E, y \in F\} \quad \text{if } d(E,F) = 0 \\ \forall \varepsilon > 0 \ \exists (x,y) \in E \times F, \quad |x-y| < \varepsilon \end{array}}$ ✗

$\forall x \in E \quad \text{dist}(x, F) := \inf\{|x-y| : \forall y \in F\}$

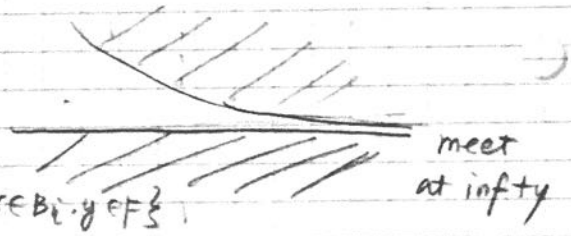
[1] $\text{dist}(x, F) > 0 \ \forall x \in E$ because if $\exists x_0 \in E \ \text{dist}(x_0, F) = 0$

it means $\forall \frac{1}{n} > 0 \ \exists y_n \in F, \quad |x - y_n| < \frac{1}{n} \Rightarrow x \in F$ contradicts $E \cap F = \phi$

[2] Because $\text{dist}(x, F) := r_x > 0 \ \forall x \in E$. $B(x, \frac{1}{2} r_x)$ is an open cover of E

$\exists B_1 \cdots B_n$ s.t. $E \subset B_1 \cup \cdots \cup B_n$

then $d(E,F) \geq \min\{d(B_1), d(B_2), \cdots, d(B_n)\}$

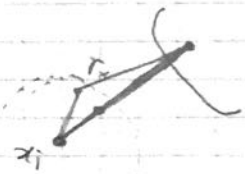


Because $\{|x-y| : x \in E, y \in F\} \subset \bigcup_{i=1}^{n} \{|x-y| : x \in B_i, y \in F\}$

so $d(E,F) \geq \inf \bigcup_{i=1}^{n} \{|x-y| : x \in B_i, y \in F\}$

And $\inf A \cup B = \min(\inf A, \inf B)$

Because suppose $\inf A \leq \inf B$ then $\inf(A \cup B) \leq \inf A \leq \inf(A \cup B)$



$d(E,F) \geq \min_i \inf\{|x-y| : x \in B_i, y \in F\}$

then $\forall x \in B_i, \ \forall y \in F, \ r_{x_i} \leq |x_i - y| \leq |x_i - x| + |x - y|$

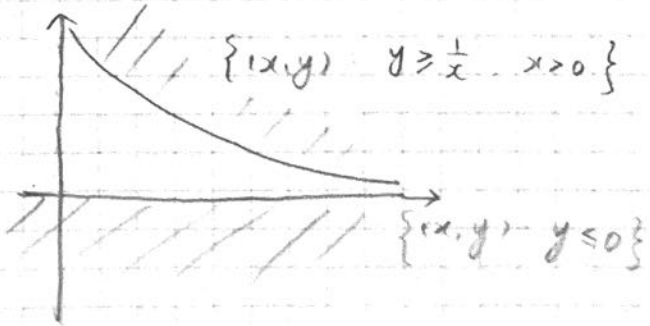
$\Rightarrow |x-y| \geq \frac{1}{2} r_{x_i}$

Hence $\frac{1}{2} r_{x_i}$ is a lower bound

$\inf\{|x-y| : x \in B_i, y \in F\} \geq \frac{1}{2} r_{x_i} > 0$

$\Rightarrow d(E,F) > 0$ $\square$

Counter Example



1.3 Leb Measurability

Def. $E \subset \mathbb{R}^d$ is Leb measurable if $\forall \varepsilon > 0$, $\exists$ open $U \subset \mathbb{R}^d$, $E \subset U$ and $m^*(U \setminus E) \leq \varepsilon$ $\square$

Lemma. Every open, closed subsets of $\mathbb{R}^d$ is Leb-measurable, etc. Every Jordan meas. sets are Leb-meas. $\square$

Prop. Criteria for Measurability, $E \subset \mathbb{R}^d$ is meas. iff

(1) $\forall \varepsilon > 0$, $\exists$ open U, $m^*(U \setminus E) \leq \varepsilon$, $U \supset E$

(2) $\forall \varepsilon > 0$, $\exists$ open U, $m^*(U \triangle E) \leq \varepsilon$

(3) $\forall \varepsilon > 0$ $\exists$ closed F, $m^*(E \setminus F) \leq \varepsilon$ $F \subset E$

(4) $\forall \varepsilon > 0$ $\exists$ closed F, $m^*(E \triangle F) \leq \varepsilon$

(5) $\forall \varepsilon > 0$ $\exists$ Leb measurable set E_ε s.t. $m^*(E \triangle E_\varepsilon) \leq \varepsilon$ $\square$

Thm. $m(\phi) = 0$ and $m(\bigsqcup_{n=1}^{\infty} E_n) = \sum m(E_n)$ for measurable (E_n)

Rmk. This is a significant improvement over Leb outer measure.

proof. [Prop (3)] Let E be measurable

If F is closed, then $\mathrm{int}(F) =: \mathring{F}$ is open and $m^*(F \setminus \mathring{F}) = m^*(\partial F)$

3. Lebesgue integral

$\int_{\mathbb{R}^d} \cdot \, d\mu$ is a map (functional) $\mathcal{M}^+ \to [0,+\infty]$

or $L^1 \to \mathbb{R}$

where $\mathcal{M}^+$ is the set $\{f : \mathbb{R}^d \to [0,+\infty],\ f \text{ is measurable}\}$

L^1 is the set $\{f : \mathbb{R}^d \to \mathbb{R},\ \int |f| < \infty\}$ $\square$

3.1 Measurable functions $\mathcal{M}^+$

Def. f^{-1} (measurable B) is measurable

Equiv. f is pointwise limit of simple functions

proof. If $f : (X, \mathcal{M}) \to \mathbb{R}$ is measurable, then the usual $(f_n) \to f$ poinduise $\square$

<u>prop.</u> $f, g \in \mathcal{M}^+,\ a \geq 0 \Rightarrow f+g \in \mathcal{M}^+ \quad af \geq \mathcal{M}^+$

$fg \in \mathcal{M}^+$

prop. $(f_n) \in \mathcal{M}^+$, $\sup f_n$ $\inf f_n$ $\limsup f_n$ $\liminf f_n \in \mathcal{M}^+$

proof $\sup f_n(x) \leq c \Rightarrow \forall n,\ f_n(x) \leq c$, so $(\sup f_n)^{-1}[0,c] = \bigcap_{n=1}^{\infty} f_n^{-1}[0,c]$

$\inf f_n(x) \leq c \Rightarrow \exists n$ s.t. $f_n(x) \leq c$ so $(\inf f_n)^{-1}[0,c] = \bigcup f_n^{-1}(c,+\infty)$

prop. f continuous $\Rightarrow f$ is measurable

3.2 Markov Inequality

$f : \mathbb{R}^d \to [0,+\infty]$ measurable, $\forall\ 0 < \lambda < \infty$

$m\{x : f(x) \geq \lambda\} \leq \frac{1}{\lambda} \int_{\mathbb{R}^d} f(x)\,dx$

<u>Rmk.</u> First moment method

1.3.5 Littlewood's Three Principles

(1) Every measurable set is nearly a finite sum of intervals

(2) Every function in L^1 is nearly continuous

(3) Every pointwise convergent sequence of functions is nearly uniformly convergent.

1. First Principle

A measurable set can be outer approximated by an open set.
inner approx. by a closed set

2. Second Principle

Thm. Let $f \in L^1(\mathbb{R}^d)$, $\varepsilon > 0$, then

(1) $\exists$ absolutely integrable function g such that $\|f-g\|_{L^1} \leq \varepsilon$

(2) $\exists$ step function g such that $\|f-g\|_{L^1} \leq \varepsilon$

(3) $\exists g \in C_c(\mathbb{R}^d)$ (continuous, compactly supported) $\|f-g\|_{L^1} \leq \varepsilon$

proof. (1) $\exists$ simple func. $g \leq f$, $\int g \geq \int f - \varepsilon$ if $f \in \mathcal{M}^+$

then $\int f - g \leq \varepsilon$

If $f \in L^1$, then $f^+, f^- \in \mathcal{M}^+$ $\exists g_1, g_2$

$\int |f^+ - g_1| \leq \varepsilon/2$ $\int |f^- - g_2| \leq \varepsilon/2$

$\int |g_1 - g_2| \leq \int g_1 + \int g_2 < \infty$ so $g_1 - g_2 \in L^1$

and $\|f^+ - f^- - (g_1 - g_2)\| \leq \varepsilon$

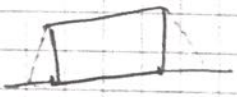
(2) Step functions are defined as $\sum^n c_j 1_{B_j}$ where B_j are boxes

Going from a simple func $f_1 \in L^1$ s.t. $\|f - f_1\|_{L^1} \leq \frac{\varepsilon}{2}$

Then $f_1 = \sum_{j=1}^n a_j E_j$ Let $f_{2,j} = a_j E_j$

Then $f_1 = \sum_{j=1}^n f_{2,j}$ Let $g_j = a_j B_j$ s.t. $m(B_j \Delta E_j) < \frac{\varepsilon}{n|a_j|}$

Then $\|f_1 - \sum g_j\|$

(3) Directly construct for a box 

Remark. Extention to smooth compactly supported functions: test functions

3. Third Principle

Def. (1) pointwise convergence $f_n(x) \to f(x)$ $\forall x$

(2) a.e. pointwise

(3) uniform convergence $\forall \varepsilon > 0, \exists N$ s.t. $|f_n(x) - f(x)| \le \varepsilon$ $\forall n > N, \forall x \in \mathbb{R}^d$

Def. Locally uniform conv. f_n $\mathbb{R}^d \to \mathbb{C}$ $\to f$ locally uniformly

if $\forall$ ~~bounded subset~~ open neighborhood of x $E \subset \mathbb{R}^d$, $f_n \rightrightarrows f$ on the neighborhood.

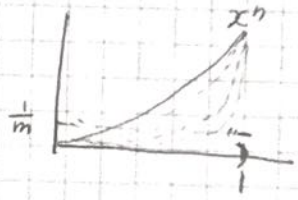
Thm. Egorov's theorem. Let f_n $\mathbb{R}^d \to \mathbb{C}$ be a sequence of measurable functions that a.e. pointwise converges to f. $\forall \varepsilon > 0$

$\exists$ leb measurable set A of measure $\le \varepsilon$, s.t. $f_n \to f$ locally uniformly to f on A^c.

proof. (1) ~~Assume $f_n(x) \to f(x)$ $\forall x \in \mathbb{R}^d$, then $\forall \delta > 0$ $\exists N$ s.t. $\forall n \ge N$ $|f_n(x) - f(x)| < \delta$~~

~~Fix δ $\bigcup_N \bigcap_{n \ge N} E_\delta = \emptyset$ $E_\delta = \{x : |f_n(x) - f(x)| > \delta\}$~~

$E_{N,m} = \{x : |f_n(x) - f(x)| > \frac{1}{m}$ for some $n \ge N\}$



Then $\bigcap_N E_{N,m} = \emptyset$ $\forall m$ $\forall R > 0$

$\lim_N m(E_{N,m} \cap B(0,R)) = 0$

so $\forall m \ge 1$ $\exists N_m$ s.t. $m(E_{N,m} \cap B(0,m)) \le \frac{\varepsilon}{2^m}$ $\forall N \ge N_m$

$A = \bigcup_{m=1}^{\infty} E_{N_m, m} \cap B(0,m)$ $m(A) \le \varepsilon$

1.5.2 Modes of convergence in terms of step functions

$f_n = A_n 1_{E_n}$, controlled by (1) height A_n of f_n

(2) width $\mu(E_n)$ of f_n

(3) N^{th} tail support $E_N^* = \bigcup_{n \geq N} E_N$

prop. (1) $f_n \rightrightarrows 0$ iff $A_n \to 0$ as $n \to \infty$

(2) $f_n \xrightarrow{L^\infty} 0$ iff $A_n \to 0$ (for step functions)

(3) $f_n \overset{\text{almost}}{\rightrightarrows} 0$ iff $A_n \to 0$ or $\mu(E_N^*) \to 0$ as $N \to \infty$

proof: $\Rightarrow$ If $A_n \to 0$ $f_n \rightrightarrows 0$ otherwise $\forall \varepsilon > 0$ $\exists E$ s.t. $\mu(E) \leq \varepsilon$.

and on $X \setminus E$. $A_n \to 0$

Define $E_N = \{ \bigcup_{n \geq N} E_n \}$ then $E_N \subset E \cup A_n$

on $X \setminus E$. $\forall \varepsilon > 0$ $\exists N$, s.t.

(3) $\forall \varepsilon > 0$ ~~$\exists$ s.t~~ $\exists E$. s.t $\mu(E) < \varepsilon$, on $X \setminus E$. $f_n \rightrightarrows 0$ $f_n = A_n 1_{E_n}$

~~So $\forall \varepsilon > 0$ $\exists E$ s.t~~ if $\mu(E_N^*) \not\to 0$. then $\liminf \mu(E_N^*) = c > 0$

$\exists m$ $\inf_{n \geq m} \mu(E_n^*) > c - \delta$ $\forall m$ $\mu(E_n^*) > c\delta$

$\forall n \geq m$ $\mu(E_n^*) > 0$

then f_n can't $\overset{\text{almost}}{\rightrightarrows}$ 0

$\Leftarrow$ If $\mu(E_N^*) \to 0$ then $\forall \varepsilon > 0$ $\exists N_1$ $\forall N \geq N_1$ $\mu(E_N^*) < \varepsilon$

then let $E = \bigcup E_N^*$ on E^c $f_n = 0$ $\forall n \geq N$

4) $f_n \to 0$ iff $A_n \to 0$ or $\bigcap_{N=1}^{\infty} E_N^* = \emptyset$

Excercise 1.5.5. $\sum \|f_n - f\|_{L_1} < \infty$ in terms of $A_n 1_{E_n}$

$\sum A_n \mu(E_n) < \infty$ $\lim_n A_n \mu(E_n) \Rightarrow 0$

If $A_n \to 0$ otherwise $A_n \geq c$ $0 \leftarrow \lim A_n \mu E_n \geq \lim c \mu E_n$

$\lim \mu(E_n) \to 0$

$\sum A_n \mu E_n \to 0$ hence

$\sum_{n=N}^{M} A_n \mu E_n < \varepsilon$ $\quad \forall \varepsilon > 0$ Cauchy $\exists N \ \forall n, M \geq N$ then if $A_n \not\to 0$ because it's positive $\exists c > 0$ st $\overset{\liminf}{A_n} \geq c$

$A_n \not\to 0$ $\quad \limsup A_n \geq c$ $\quad \limsup A_n = \infty$

$\mu E_n \to 0$

$$\sum_{n=N}^{M} A_n \mu E_n \geq \sum_{n=N}^{M} \mu\left(\bigcup_{n=N}^{M} E_n\right) \sum_{n=N}^{M} A_n$$

$\sum \|f_n - f\|_{L_1} < \infty$ then $f_n \rightrightarrows f$ almost uniformly

$\sum A_n \mu(E_n) < \infty$ $\quad$ then $T_N = \sum_{n=N}^{\infty} A_n \mu(E_n) \to 0$

$\langle a | b \rangle \leq \|a\| \|b\|$

$$\sum_N A_n \mu(E_n) \leq \varepsilon \leq \sqrt{\sum A_n^2 \sum (\mu(E_n))^2} \qquad \langle A_n | \mu(E_n) \rangle$$

$A_n \not\to 0$

$\exists$ subsequence

A_n bounded

$$\geq \sum_N A_n \mu\left(\bigcap_N E_n\right) \geq \sum_{n \geq N} \inf A_n \ \mu\left(\bigcap_N E_n\right)$$

$$\geq A_m \mu\left(\bigcap_N E_n\right) \qquad \exists m \text{ s.t. } A_m > 0$$

Hence $\mu\left(\bigcap_N E_n\right) \to 0$ $\quad$ Hence $\cancel{\lim \mu E_n = 0}$

$\cap$

$A_n \mu E_n \to 0$

$A_n \mu \cap E_n \to 0$

$$\mu\left(\bigcap_N E_n\right) \leq \frac{A_m}{A_m} \mu(E_n) \leq \frac{\sum A_n \mu E_n}{A_m} \to 0$$